

# Sreevatsa RAVIKANTH, MSc in Bioinformatics

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## Employment History

### Technical experience

- Apr 2026 – Present    📌 **Full Stack Engineer**, Districom, Montigny-lès-Cormeilles, France.  
- Development of a mobile application for retail stock management for Sony France  
- Retail product promotion and technical demonstration of Sony products
- Apr 2025 – Feb 2026    📌 **Webmaster & Research Coordination**, nutritionDay worldwide, Villejuif, France.  
- Creation and management of the French branch website for the international nutritionDay study  
- Adaptation of an international clinical research project to the French healthcare context  
- Technical maintenance, user support, and coordination of national participation in a global epidemiological study
- Mar 2025 – Oct 2025    📌 **Junior biostatistician**, Gustave Roussy, Villejuif, France.  
- Contribution to scientific writing (second author of a manuscript under review)  
- Statistical study on micronutrient deficiencies in head and neck cancer patients  
- Construction and structuring of clinical databases for predictive modeling  
- Univariate + multivariate and LASSO logistic regressions
- Apr 2024 – Sep 2024    📌 **Junior biostatistician**, Gustave Roussy, Chevilly-Larue, France.  
- Co-authorship of a scientific manuscript under review  
- Data preprocessing and statistical analysis of 204 oncology patients  
- Clinical data structuring for research in oncology nutrition - Calculation of prevalences and subgroup comparisons using Boschloo's exact tests - Identification of 6 significant micronutrients related to inflammation through Spearman correlations and symbolic regression modeling (95% CI)
- Jan 2023 – Feb 2023    📌 **Process Optimization Consultant**, D-MUTE NETWORK, Saint-Maurice, France.  
- Design and development of APIs for enterprise resource management  
- Integration of Microsoft 365 licensing data into SharePoint systems

### Other experience

- May 2018 – Present    📌 **Elderly Care Assistant**, URSSAF Île-de-France, Joinville-le-Pont, France.
- Apr 2026 – Present    📌 **Product Promoter**, Sony France / Districom, Montigny-lès-Cormeilles, France.
- Dec 2021 – Aug 2022    📌 **Sales Advisor**, Xiaomi France, Paris, France.
- Feb 2016 – Apr 2016    📌 **Animator**, Youth and Archive Department, City of Joinville-le-Pont, France.

## Education

- 2020 – 2025     **Engineering Degree (MSc equivalent), École Centrale d'Électronique Paris**, in Computer Science & Electronics, specialization Health & Technology.  
Master's Thesis: *Rapport de stage de fin d'études*.  
International exchanges: *University of Verona (2024)*, *OMNES London School (2022)*.
- 2017 – 2020     **Scientific Baccalaureate, Lycée Teilhard de Chardin**, Specialization Mathematics & Earth and Life Sciences (SVT), with high honours.

## Research Publications

### Journal Articles

- 1 M. Guinhut et al., "Prevalence of micronutrient deficiencies in 465 patients with head and neck cancers (hnc) admitted to a rehabilitation unit," *Clinical Nutrition ESPEN*, 2025, Manuscript under review.
- 2 A. Konesalingam, S. Ravikanth, G. M. Machado, F. Belbachir, and A. Jaouadi, "Stress detection with physiological signal from wrist-worn devices," *International Journal of Medical Informatics*, 2025, Manuscript under review.

## Research Projects

- 2023 – 2025     **Stress Detection Using Machine Learning**, LyRIDS, Paris, France  
Co-lead author of a scientific manuscript currently under review
- Harmonization of multimodal physiological datasets (PPG, TEMP, EDA, ACC)
  - Advanced signal preprocessing: filtering, temporal and frequency feature engineering, dimensionality reduction
  - Analysis of multimodality impact and windowing strategies
  - Training and evaluation of 12 machine learning models using LOSO-CV
  - Bayesian hyperparameter optimization
  - Best performance: F1-score of 81.9% (ternary classification)
  - Model interpretability using SHAP values
  - Development of reproducible ML pipelines on Linux systems
- 2020 – 2022     **ECE CUP**, LyRIDS, Paris, France  
Semi-finals, ranking among the top teams out of more than 300 competing teams.
- Developed an autonomous robotics platform using a ESP32.
  - Implemented real-time image processing with OpenCV to localize the robot, barrels, and opponent on the playing field.
  - Designed path-planning and obstacle-avoidance algorithms for autonomous navigation.
  - Integrated wireless communication through MQTT for monitoring and control.
  - Competed in a robotics challenge involving barrel collection and delivery while avoiding collisions with opponents and obstacles.

## Skills

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|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages             |  French (native), Tamil (native), English (C2, TOEIC 980/990), Japanese (A1), Spanish (A1)                                                                                                                           |
| Programming Languages |  Python, Java, C/C++, SQL, R, MATLAB/Octave, Dart/Flutter, SAS, VHDL                                                                                                                                                 |
| Data Science & AI     |  Bioinformatics, Machine Learning, Deep Learning, Supervised Learning, Image Processing, Reinforcement Learning, Bayesian Optimization, Signal Processing, Stress Detection, Gene Expression Profiling, Graph Theory |

## Skills (continued)

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|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Web Development     | ■ HTML, CSS, JavaScript, PHP, MySQL, Wordpress, Hostinger                                                                                                                                                 |
| Tools & Software    | ■ Microsoft Office (Word, Excel, PowerPoint), Microsoft Power Platform (PowerApps, Power Automate, Power BI), SharePoint, L <sup>A</sup> T <sub>E</sub> X, Canva, Adobe Photoshop, SolidWorks, ThinkerCAD |
| Clinical & Research | ■ Good Clinical Practice (ICH E6 R2), Compliance Regulations, Intellectual Property                                                                                                                       |
| Project Management  | ■ Agile / Decisioning, Consulting, Marketing, Communication                                                                                                                                               |
| Embedded Systems    | ■ Embedded Linux Systems, Arduino, VHDL, Proteus, Quartus, Cisco Packet Tracer, Wireshark                                                                                                                 |

## Miscellaneous Experience

### Awards and Achievements

- 2024 ■ 1st place out of 567 – Best Scientific Poster.
- 2nd place out of 38 – Inetum Hackathon.
- 2022 ■ 4th place out of 300 – ECE Cup Robotics Competition.

### Certifications

- 2026 ■ ICH E6 (R3) Good Clinical Practice, The Global Health Network.
- 2023 ■ ECE Python Exam, Liora.
- Project Management MOOC, Centrale Lille.